Lecture 31.11: What is a Dendrogram — Structure & Cutting Thresholds











📌 Topic Overview

Detailed study of Dendrograms in hierarchical clustering. Explains tree diagram anatomy, height as Euclidean linkage distance, determining optimal K by identifying the longest unbroken vertical branch, and cutting horizontally.

📋 Key Learning Objectives

- Formal definition of a Dendrogram: Tree-based visual representation of cluster merges.
- Anatomy of the Dendrogram: X-axis (sample indices) and Y-axis (linkage merge distance).
- Determining optimal K: Finding the longest vertical line uncrossed by horizontal splits.
- Setting a horizontal distance threshold to cut the dendrogram into K discrete clusters.
- Interpreting sub-cluster hierarchies and biological taxonomy similarities.

🔍 In-Depth Conceptual Breakdown

- **Dendrogram Kya Hai?**: Dendrogram ek special tree-like diagram hota hai jo yeh record karta hai ki kaun se do data points kis distance (height) par aapas mein merge huye.

- **Anatomy Of The Plot**:

- - **X-axis**: Saare individual data points (samples) hote hain.
- - **Y-axis (Height)**: Euclidean distance ya linkage distance hoti hai. Do branches jitni upar jakar judti hain, unke beech ka distance utna hi zyada tha!

- **Optimal K Kaise Chunein? (The Golden Rule)**:

- 1. Dendrogram mein wo **sabse lambi vertical line** dhoondo jiske beech se koi bhi horizontal merge line pass na ho rahi ho.
- 2. Us vertical line ke beech se ek **Horizontal Cut Line** (threshold distance) kheench do.
- 3. Yeh cut line jitni vertical branches ko kaategi, wahi hamara **Optimal K (Number of Clusters)** hoga!

- **Ma'am Ka Blackboard Example**: Ma'am ne board par 5 samples (A, B, C, D, E) ka dendrogram draw karke dikhaya ki kaise ek horizontal cut lagane par 2 distinct major clusters ban jaate hain.

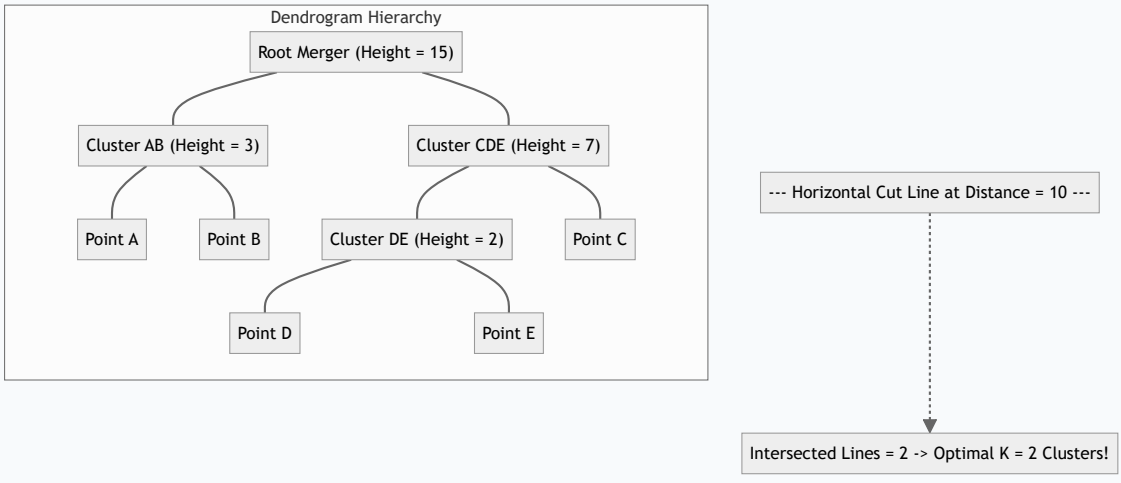
- **Pure Theory Lecture**: Is lecture mein ma'am ne dendrogram ki anatomy aur cut threshold board par draw karke sikhaya hai, koi code nahi hai.

📋 Dendrogram Components & Interpretation

Visual Element	Graphical Meaning	Mathematical Representation
Leaves (Bottom)	Individual raw observations	Singletons $S_i = \{\mathbf{x}_i\}$ at distance = 0
Vertical Stems	Clusters awaiting subsequent merger	Height proportional to linkage distance $d(A, B)$
Horizontal Bars	Merger event between two clusters	The exact dissimilarity threshold of the fusion
Cut Line (Threshold)	Horizontal partition boundary	Divides tree into K disjoint subtrees

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📊 Ma'am's Blackboard Dendrogram & Cut Threshold



🔢 Mathematical Formulation

Optimal number of clusters K given distance cut threshold θ:

$$K(\theta) = \sum_{j \in \text{Active Nodes}} \mathbb{I}\left(\text{height}(j) > \theta \geq \text{height}(\text{children}(j))\right)$$

📌 Note on Code

This is a pure conceptual, mathematical, and blackboard theory lecture. In accordance with course guidelines, no synthetic code is appended.